

Skills

- 3+ years experience data analysis, building reproducible analytical workflows, visualization and stakeholder oriented reporting.
- Experience working within a Linux environment and HPC environment (supercomputer).
- Experiences working in collaborative teams, strong problem-solving and critical thinking skills, attentive to details.
- Languages: English, Cantonese, Mandarin
- Tools: SQL, R, Python (Pandas, NumPy, Matplotlib, PyMC, Scikit-learn, Plotly), Excel, Tableau, Power BI, Git Version Control, QGIS

Education

- Sep 2025 – **MSc in Statistics**, *University of Toronto*, Toronto, ON, Canada
Apr 2026 (expected) Courses: Applied Multivariate Analysis, Methods of Applied Statistics, Special Topics in Data Science, Survival Analysis
- Sep 2021 – **H.B.Sc in Statistical Science (Specialist), Mathematics (Major), Health Studies (Focus)**, *University of Toronto*, Toronto, ON, Canada **cGPA: 3.94/4.00**
Apr 2025 Courses: Applied Statistical Methods, Applied Spatial Analysis, Time Series Analysis, Applied Bayesian Statistics, Surveys, Sampling and Observational Data, Statistical Consultation
Awards: Samuel Beatty In-course Scholarship (2025), University of Toronto Excellence Award (2024), Eleventh Annual Canadian Statistics Student Conference Undergraduate Level First Prize (2023), University of Toronto Dean's List Scholar (2021–2025)

Professional Experiences

- Sep 2025 – **R-Based Learning Designer**, *University of Toronto Human Biology Program*
Current
 - Built 4 auto-testable R assignments for a cohort of 400 students, cutting manual grading workload by 90% and accelerating feedback turnaround.
 - Designed 2 interactive LearnR modules using RShiny, reducing student debugging requests and enhancing self-directed learning efficiency.
 - Standardized version-controlled releases on Git across a multi-developer team, delivering zero student-reported issues and significantly improving instructional reliability.
- Sep 2024 – **Data Visualization Analyst**, *University of Toronto Human Biology Program*
Apr 2025
 - Analyzed learning resource engagement data for 1,000+ course enrollments, identifying behavioral patterns linked to final grades and informing assessment redesign.
 - Developed 100+ Tableau dashboards visualizing key performance metrics (discussion board access, assignment completion, early-term quiz grades), enabling instructors to identify at-risk students by week 3.
 - Translated analytical findings into data-driven recommendations that improved assessment design and feedback practices, enabling instructors to address early-term performance gaps and refine course delivery pacing.
- Jun 2024 – **Quantitative Research Analyst**, *Centre for Addiction and Mental Health*
Apr 2025
 - Analyzed how physical movement influenced cybersickness across 2 immersive VR medical educational programs, contributing to 1 peer-reviewed publication advancing VR education research.
 - Validated 90+ survey responses and conducted descriptive and non-parametric analyses, identifying a significant positive relationship between movement intensity and nausea severity.
 - Delivered data-driven recommendations on movement thresholds and user comfort, informing VR training design decisions to balance educational effectiveness with learner well-being.
- May 2023 – **Data Analyst Intern**, *Canadian Urban Institute*
– Aug 2023
 - Built regression models to analyze nationwide household carbon-emission patterns, identifying key drivers of high-emission regions and guiding targeted interventions for a national sustainable city-planning project.
 - Designed an interactive scenario-modeling tool illustrating how urbanization and housing composition influence greenhouse-gas emissions, enabling stakeholders to test policy options on a dashboard.
 - Produced 300+ geospatial maps in QGIS and integrated them with statistical findings, uncovering location-specific emission drivers and delivering actionable planning recommendations.

- Oct 2022 – **Computational Social Science Researcher**, *University of Toronto Population Well-being Lab*
- May 2023
- Applied machine learning methods, Gaussian mixture model, k-means, and hierarchical clustering (mclust, stats) to 150K+ survey observations, uncovering subpopulations based on life satisfaction and demographics.
 - Created 4 algorithm tip-sheets explaining statistical methods in accessible terms, improving onboarding and methodological literacy for lab members.
 - Designed a poster discussing clustering methods workflow and implications of results, resulting in a conference poster first prize.
- Sep 2022 – **Statistical Consultant**, *University of Toronto Data Sciences Café*
- Dec 2023
- Provided on-demand statistical guidance to 20+ interdisciplinary researchers, transforming broad research questions into statistically rigorous analysis plans.
 - Diagnosed design limitations and recommended tailored analytical approaches, helping researchers avoid common biases and strengthen study validity.
 - Authored 5 reports on advanced statistical topics, translating complex concepts and coding methods into clear guidance that enhanced statistical literacy among non-statistician researchers.

Statistics Course Projects

- Oct 2025 – **Uncovering Calendar-Based Structures Through Clustering of Daily Traffic Data**, *STA2005*
- Dec 2025
- Applied PCA and UMAP to a year-long traffic dataset across 26 locations, uncovering recurring temporal clusters tied to school calendars, holidays, and seasonal routines.
 - Created visualizations and prepared a report that informed transportation planning and infrastructure maintenance decisions.
- Oct 2025 – **Fatal US School Shootings (1999-2025)**, *STA4101*
- Dec 2025
- Modeled fatality risk in 400+ US school shootings using logistic regression, highlighting actionable risk indicators including shooting intent, absence of non-shooter injuries, and shooter connection to the school community.
 - Built a reproducible GitHub repository documenting all data cleaning, modeling, and analysis steps to support transparency and collaboration.
 - Synthesized findings into a report informing safety and emergency preparedness strategies.
- Sep 2025 – **Statistical Inference for Conditional Spatiotemporal Correlation in Brain MRI Data**, *STA4000*
- Dec 2025
- Pioneered analysis framework for assessing genetic heritability of intermodal coupling using multi-modal brain MRI data, generating methodological insights for advanced neuroimaging inference.
 - Conducted 50+ HPC simulation experiments and applied adaptive clustering techniques with parallel computing to validate proposed model's Type I error control and statistical power.
 - Summarized findings in a report that informed methodological refinements for robust neuroimaging statistical approaches.
- Feb 2025 – **Modeling Cocoa Futures Prices: A Statistical Analysis of Ghana's Cocoa Production**, *STA457*
- Apr 2025
- Integrated 3,700+ daily observations across four datasets, performing extensive cleaning (imputation, correction of invalid entries, aggregation) to produce an analysis-ready dataset.
 - Developed ARIMA and Random Forest forecasting models in R (astsa, forecast, randomForest), engineering lagged features and covariates to capture linear and nonlinear drivers of price volatility.
 - Conducted full diagnostics (ACF/PACF, Ljung-Box test, residual analysis, feature importance assessment) and authored a comprehensive statistical report on climate and economic influences on commodity pricing.
- Oct 2024 – **Transit and Crime: Investigating Assault Clusters Around Toronto's TTC Subway Lines**, *STA465*
- Dec 2024
- Analyzed 200k+ Toronto assault incidents using kernel density estimation, point process models and HDB-SCAN clustering to identify assault concentration clusters within varying buffer distances around TTC subway infrastructure.
 - Developed interactive geospatial report with maps translating statistical findings into actionable insights for urban planners and policymakers on transit-related public safety.
- Sep 2024 – **Quercus Engagement as a Predictor of First-Year University Student Dropout Risk: A Logistic Regression Analysis**, *STA490*
- Apr 2025
- Preprocessed engagement data from 3,000+ students, resolving missing values and standardizing predictors across multiple time windows.
 - Built and validated logistic regression models to assess the relationship between Quercus engagement and dropout risk, identifying early disengagement in first 4 weeks as a critical retention indicator.
 - Delivered stakeholder-ready reports with clear visualizations and accessible interpretations, informing refinement of current learning support strategies.

Sep 2024 – **Fair Machine Learning in Healthcare, STA496**

- Apr 2025
- Developed R/Python data pipeline (dplyr, Pandas, PyTorch) on 40,000+ EHR records from MIMIC-III database, producing a clean, standardized dataset that improved downstream model stability and interpretability.
 - Executed PMI-SVD embedding-generation pipelines and performed KESER feature selection on 6,000+ clinical concepts for robust prediction tasks.
 - Trained regression and random forest models in R for depression diagnosis and conducted fairness evaluation, uncovering demographic disparities driven by label bias.
 - Built a fully reproducible data workflow and curated a GitHub repository, enabling seamless replication and improving research transparency.

May 2024 **Aspect of Robust Regression Analysis, University of Toronto Department of Statistical Sciences**

- Current
- Developed and evaluated 4 parameter-estimation approaches for high-dimensional, heavy-tailed data, producing methodological insights that improved estimation accuracy over traditional regression methods.
 - Implemented and optimized algorithms in R/Python, resolving non-convergence issues and increasing computational efficiency for large-scale simulations.
 - Delivered a departmental research talk and released an R package with 20+ reusable functions, strengthening reproducibility and statistical communication.

Feb 2024 – **Investigating The Factors Affecting Cycling Accident Severity: A Bayesian Hierarchical Mixture Model Approach, STA365**

- Apr 2024
- Conducted EDA on 800K+ accident records in Python (Pandas, NumPy, Matplotlib, Seaborn), identifying key temporal and environmental predictors of severe injuries.
 - Engineered a Bayesian missing-data imputation pipeline (Scipy) using posterior predictive sampling, restoring 60K+ incomplete records with validated convergence diagnostics.
 - Built a Bayesian hierarchical mixture model in PyMC incorporating time-of-day structure, spike-and-slab selection, and demographic mixture components, achieving measurable performance gains over baseline logistic regression.

Sep 2023 – **An Agent-Based Simulation Approach to Investigate the Effect of Decreasing Birth Rates and the Efficacy of Potential Solutions, ROP399**

- Apr 2024
- Designed 20+ agent-based simulations modeling population dynamics and evaluating policy interventions for declining birth rates, identifying two high-impact levers that improved long-term population stability.
 - Synthesized 30+ multidisciplinary sources into an annotated bibliography, strengthening model assumptions and improving simulation realism.
 - Conducted time-series and regression analyses in Python (SciPy) to quantify intervention effects, revealing short- vs long-term trade-offs that informed policy recommendations, resulting in a poster presentation.

Health Studies Course Projects

Feb 2024 – **Effectiveness of Usage of Virtual Reality in Raising Awareness about Cannabis Misuse Issues among Teenages, HMB342**

- Apr 2024
- Designed a randomized controlled trial study comparing virtual reality-based education with traditional seminars to evaluate effectiveness in raising awareness of cannabis misuse among Canadian high school students.
 - Developed evaluation metrics and statistical analysis plan (pre/post surveys, T-tests, knowledge retention, behavioral intentions) to measure intervention impact on awareness and future cannabis use.
 - Applied health promotion and program design frameworks to integrate emerging technologies into adolescent substance misuse education, highlighting cost-effectiveness and policy implications for public health strategies.

Feb 2023 – **Food Insecurity Intervention Program Planning, HST330**

- Apr 2023
- Conducted root cause analysis and developed a program proposal to address food insecurity among Toronto low-income families, modeling intervention design on the Supplemental Nutrition Assistance Program with a digital e-coupon system.
 - Designed a quasi-experimental evaluation plan, including baseline and post-intervention measures (Food Insecurity Experience Scale, BMI, Healthy Eating Index, stress levels) to assess program effectiveness.
 - Created a logic model and theory of change framework to outline expected outcomes, health implications, and long-term policy applications, demonstrating strong skills in program evaluation and health policy analysis.

Oct 2022 – **Proposal to Address Teenager Substance Misuse Problem, HST209**

- Dec 2022
- Conducted comprehensive analysis of risk factors (family influence, peer pressure, media exposure, mental health) contributing to teenage substance misuse, synthesizing evidence-based research into actionable insights.
 - Designed and evaluated multi-level interventions including individual counseling, family therapy, in-school workshops, and peer-led support communities to promote adolescent health and reduce substance misuse.
 - Developed health promotion strategies grounded in empowerment and community engagement, aligning interventions with public health models to improve long-term outcomes for at-risk youth.

Other Experiences

- Sep 2023 – **Teaching Assistant and Learning Support**, *University of Toronto Department of Statistical Sciences and Data Sciences Institute*
- Led tutorials and office hours across multiple statistics courses in person and online, strengthening undergraduate students and industry professionals understanding of R, Python, SQL, and statistical concepts.
 - Supported high-school outreach events, Florence Nightingale Day, translating advanced statistical ideas into engaging, accessible explanations.
- Sep 2022 – **First-year Learning Community Assistant Peer Mentor**, *University of Toronto*
- Apr 2023
- Facilitated 12 sessions supporting 20+ first-year students in academic planning, study strategies, and transition management.
 - Built an inclusive learning community and coordinated discussions with faculty and library staff to enhance student engagement.
 - Managed the online Quercus course page to facilitate remote communication and resource access.
- May 2022 **Bilingual Client Care Representative**, *Telus Health*
- Sep 2022
- Served as first point of contact for 100+ daily client inquiries, delivering bilingual (English/Chinese) support that improved satisfaction and reduced escalations.
 - Assessed client needs, identified the purpose of each inquiry, and provided tailored service recommendations, ensuring clients felt informed, supported, and guided toward the right solutions.
 - Coordinated service delivery by booking appointments and routing callers to the appropriate service-specific teams, contributing to faster resolution times and smoother operations.